

Decision Making Algorithm for REPEAT (≤ 72 hrs since prior) Blood Cultures (BC)

START

Does the patient have:

Positive blood cultures (BC) ❶ with **ONE** of the following:

- *Staph aureus*
- *Staph lugdunensis*
- *Candida sp.*
- Gram negative rods in setting of IVDU
- Enterococcus sp.
- Viridans group Streptococcus
- β -hemolytic *Streptococcus*
- Any organism associated with suspected endovascular infection ❷

No

Diagnosed with a central-line associated bloodstream infection (CLABSI) in the last 72 hours **AND** retained catheter?

Yes

• Repeat 2 sets of peripheral BC daily from 2 different sites until negative for 48 hours
• Recommend Infectious Diseases consult

No

Persistent fever or leukocytosis **AND** 2 sets of negative BC within 48 hours while other sources are being evaluated.

Yes

No

Clinical response after starting antibiotics and presumed source control and no concern for endovascular infection

Yes

No

Cultures likely to represent contamination (e.g. single BC with diphtheroids, *Cutibacterium sp.*, coag-neg *Staph*, micrococci, *Bacillus sp.*, lactobacilli, veillonella, peptostreptococci)

Yes

No

Still concern for infectious cause?

Yes

Consider Infectious Diseases consult

Repeat blood cultures not indicated if <72 hours since prior

Were two sets of BC able to be obtained?

Yes

No

Blood culture work up complete

Proceed to Difficult Blood Culture Logistics Algorithm

Definite Indication

No Indication

FOOTNOTES:

- ❶ Or if Gram Positive Cocci (GPC) on gram stain or culture, not yet speciated
- ❷ High-risk endovascular infection = Infective endocarditis (IE), septic thrombophlebitis, ICD/pacemaker in place, vascular graft, ventricular assist device (VAD), prosthetic valve or prosthetic material for valve repair, history of IE, unrepaired congenital heart disease

This guideline is not a substitute for clinical judgment. Please consider ID consultation for assistance with diagnosis and management.